US Patent & Trademark Office
Patent Public Search | Text View

United States Patent	12404738
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Rusten; Torgeir et al.

Downhole pressure equalizer

Abstract

The present invention relates to a downhole pressure equalizer comprising an elongated housing comprising a piston compartment having a first end provided in fluid communication with an outside of the elongated housing and a second end provided in fluid communication with a process chamber of a well tool. A piston device provided in the position compartment comprises an outer piston section slidingly and sealingly engaged with the piston compartment; and an inner piston section slidingly and sealingly engaged within the through bore of the outer piston section.

Inventors:	**Rusten; Torgeir (Vikhammer, NO), Sandvik; Gaute (Saksvik, NO)**
Applicant:	**Interwell P&A AS** (Hafrsfjord, NO)
Family ID:	**81940399**
Assignee:	**INTERWELL P&A AS (Røyneberg, NO)**
Appl. No.:	**18/281016**
Filed (or PCT Filed):	**May 05, 2022**
PCT No.:	**PCT/EP2022/062176**
PCT Pub. No.:	**WO2022/243056**
PCT Pub. Date:	November 24, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20240141753 A1	May. 02, 2024

Foreign Application Priority Data

NO	20210637	May. 21, 2021

Publication Classification

Int. Cl.: **E21B33/13** (20060101); **E21B23/04** (20060101); **E21B29/02** (20060101); **E21B36/00** (20060101)

U.S. Cl.:

CPC **E21B33/13** (20130101); **E21B23/0412** (20200501); **E21B23/0414** (20200501); **E21B23/042** (20200501); **E21B29/02** (20130101); **E21B36/008** (20130101);

Field of Classification Search

CPC: E21B (23/0412); E21B (23/0414); E21B (23/042); E21B (29/02); E21B (33/13); E21B (33/134); E21B (36/008)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2012/0255742	12/2011	Cortez	N/A	N/A
2013/0205888	12/2012	Teowee	102/210	F42D 5/02
2014/0014352	12/2013	Hallundbæk	166/320	E21B 43/1195
2018/0058154	12/2017	Kitchen	N/A	E21B 23/00
2018/0148991	12/2017	Hearn	N/A	H01L 25/50
2019/0178049	12/2018	Kræmer	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
20190537	12/2019	NO	N/A
2015/023807	12/2014	WO	N/A
2019/151870	12/2018	WO	N/A

OTHER PUBLICATIONS

International Search Report issued Aug. 18, 2022 in International (PCT) Application No. PCT/EP2022/062176. cited by applicant
Search Report issued Dec. 21, 2021 in Norwegian Application No. 20210637. cited by applicant

---

Primary Examiner: Coy; Nicole

Assistant Examiner: Railey; Jennifer A

Attorney, Agent or Firm: Wenderoth, Lind & Ponack, L.L.P.

Background/Summary

FIELD OF THE INVENTION
(1) The present invention relates to a downhole pressure equalizer for equalizing the pressure inside a well tool with the well pressure outside the well tool.

BACKGROUND OF THE INVENTION

(2) A pyrotechnic mixture typically comprises particulate matter, in which voids may be present. When a well tool device having a housing with a compartment containing the pyrotechnic mixture is lowered into the well, there will be a pressure difference between the well pressure outside the housing and the pressure inside the compartment of the housing. As the pyrotechnic mixture will melt the housing, there will be a sudden pressure equalization between the outside the housing and the compartment, which may impact the heat generation process negatively. In some cases, the heat generating process may stop due to such a sudden equalization.

(3) WO2013/135583 (Interwell P&A AS), it is disclosed method for performing a P&A operation wherein a first step, it was provided an amount of a pyrotechnic mixture (for example thermite) at a desired location in the well and thereafter to ignite the pyrotechnic mixture to start a heat generation process. It is also disclosed a tool for transporting the pyrotechnic mixture into the well before ignition.

(4) The transportation tool must store and protect its content until it has reached the intended position in the well. It is therefore of key importance that the tool can withstand the increasing ambient pressure exerted on it as it is lowered into the well. In the event of a collapse, the content of the tool will likely be destroyed and lost. A collapsed tool can also be difficult if not impossible to install in the well. To withstand external pressure, tools are typically made of expensive high strength materials or their wall thickness is increased which require more material which in turn increase cost.

(5) NO 20190537 describes a well tool device and a method for forming a permanent well barrier. The well tool device comprises: a housing; a movable partition device provided within the housing, the partition device separating an inner volume of the housing in a first volume defining a first compartment and a second volume defining a second compartment. A pyrotechnic mixture is provided in the first compartment; and a fluid line providing fluid communication between the second compartment and an outside of the housing. While running the well tool device into the well, the pressure difference between a pressure inside the first compartment and a pressure inside the second compartment is equalized by means of the partition device being affected by the pressure inside the second compartment.

(6) When the heat generating process is initiated, heat and gas will be produced as part of the process. This will cause the pressure inside the compartment to increase again.

(7) One object of the present invention is to improve the heat generating process.

SUMMARY OF THE INVENTION

(8) The present invention relates to a downhole pressure equalizer comprising: an elongated housing comprising a piston compartment having a first end provided in fluid communication with an outside of the elongated housing and a second end provided in fluid communication with a process chamber of a well tool; a piston device slidingly and sealingly engaged within the piston compartment, wherein the piston device is separating the piston compartment into a first sub-compartment between the piston device and the first end and a second compartment between the piston device and the second end;

wherein the piston device comprises: an outer piston section slidingly and sealingly engaged with the piston compartment; wherein the outer piston section comprises a sleeve having a through bore; an inner piston section slidingly and sealingly engaged within the through bore of the outer piston section;

wherein the downhole pressure equalizer is configured to be in one of the following states: a first state, in which the pressure difference between the fluid pressure in the first sub-compartment and the fluid pressure in the second sub-compartment causes the outer piston section and the inner piston section to move towards the second end of the piston compartment; a second state, in which the pressure difference between the fluid pressure in the first sub-compartment and the fluid pressure in the second sub-compartment causes the inner piston section to move towards the first

end of the piston compartment while the outer piston section is kept stationary.
(9) In the first state, as the outer piston section and the inner piston section are moving towards the second end of the piston compartment, the pressure difference between the pressure in the first sub-compartment and the pressure in the second sub-compartment will be reduced.
(10) Also in the second state, as the inner piston section is moving towards the first end of the piston compartment, the pressure difference between the pressure in the first sub-compartment and the pressure in the second sub-compartment will be reduced.
(11) In the first state, the pressure difference is positive, i.e. the fluid pressure in the first sub-compartment is higher than the fluid pressure in the second sub-compartment.
(12) In the second state, the pressure difference is negative, i.e. the fluid pressure in the first sub-compartment is lower than the fluid pressure in the second sub-compartment.
(13) In one aspect, the downhole pressure equalizer is configured to be in one of the following states: an initial state, in which the outer piston section and the inner piston section are located in the first end of the piston compartment and wherein the second piston compartment is pressurized with a predetermined fluid pressure.
(14) The predetermined fluid pressure is typically determined based on the expected well pressure at the well depth at which the downhole pressure equalizer and the well tool is to be used.
(15) In one aspect, the downhole pressure equalizer is configured to be in one of the following states: a final state, in which the inner piston section has slid out of its sealing engagement with the through bore of the outer piston section.
(16) In the final state, the piston device is no longer considered to be a piston, as there is a direct fluid communication between the first sub-compartment and the second sub-compartment.
(17) In one aspect, the downhole pressure equalizer comprises a longitudinal communication bore for sensing the pressure in the second sub-compartment at a location adjacent to the first end of the housing.
(18) In one aspect, the longitudinal communication bore is a fluid line, wherein a sensor for sensing the pressure in the second sub-compartment is located in the bore adjacent to the first end of the housing.
(19) In one aspect, the downhole pressure equalizer comprises: a longitudinal rod provided through the piston compartment and secured at both ends to the housing;
wherein the outer piston section is slidingly and sealingly engaged with the outer surface of the rod; and
wherein the inner piston section is slidingly and sealingly engaged with the outer surface of the rod.
(20) In one aspect, the longitudinal rod is provided centrally through the piston compartment.
(21) In one aspect, the longitudinal communication bore is provided within the rod.
(22) In one aspect, a first friction parameter representing a friction for moving the outer piston section relative to the housing is larger than a second friction parameter representing a friction for moving the inner piston section relative to the outer piston section.
(23) In one aspect, the first friction parameter and the second friction parameter are defined by the number of, and/or the properties, of sealing elements sealingly engaged between the piston device and the piston compartment and between the inner piston section and the outer piston section.
(24) In one aspect, the outer piston section comprises a sleeve in which the bore is provided, wherein the sleeve comprises a stop preventing relative movement between the inner piston section and the sleeve when the downhole pressure equalizer is in the first state.
(25) In one aspect, the inner piston section comprises a first end section, a second end section and an intermediate section, wherein the second end section is sealingly engaged within the bore of the outer piston section and where a first surface of the second end section is faced towards the first piston compartment and wherein a second surface of the second end section is faced towards the second piston compartment.
(26) In one aspect, the first end section of the inner piston section comprises a collar protruding

from the bore wherein the collar is slidingly engaged with the piston compartment.
(27) Hence, in the second state, the second end section of the inner piston section is sliding along the bore while the first end section of the inner piston section is sliding along the piston compartment.
(28) The present invention also relates to a well tool assembly for forming a permanent barrier in a well, wherein the well tool assembly comprises: a plugging and abandonment tool comprising a housing, a chamber within the housing, and a heat generating mixture and an igniter located within the chamber; a downhole equalizing tool according to any one of the above claims connected above the plugging and abandonment tool, wherein the second sub-compartment is provided in fluid communication with the chamber of the plugging and abandonment tool.
(29) In one aspect, the well tool assembly comprises: a control and logging system connected above the downhole equalizing tool, wherein the control and logging system comprises: a first pressure sensor for logging a parameter representative of a first pressure in the first sub-compartment; a second pressure sensor for logging a parameter representative of a second pressure in the second sub-compartment;
wherein control and logging system is configured to verify that the first pressure is equalized with the second pressure when the well tool assembly has been lowered to the desired location in the well and before the heat generation mixture is ignited by the igniter.
(30) In one aspect, the control and logging system is configured to verify that the first pressure is equalized with the second pressure after the heat generation process has finished.
(31) In one aspect, the first pressure equals the pressure in the well.
(32) According to invention above, it is achieved that the pressure difference between the well pressure and the internal compartment pressure is equalized before the heat generating process starts. In addition, it is achieved that the pressure difference between the well pressure and the internal compartment pressure is equalized in the initial phase of the heat generating process and further throughout the heat generating process.
(33) The term “upper”, “above”, “lower”, “below” etc. are used herein as terms relative to the well. Parts referred to as “upper” or “above” are relatively closer to the top of the well than the parts referred to as “lower” or “below”, which are relatively closer to the bottom of the well, irrespective of the well being a horizontal well, a vertical well or an inclining well.

Description

DETAILED DESCRIPTION
(1) Embodiments of the invention will now be described with reference to the enclosed drawings, wherein:
(2) FIG. **1** illustrates a cross sectional side view of the equalizing tool in the initial state, where the piston device is in the first end of the piston compartment;
(3) FIG. **2** illustrates a cross sectional side view of the equalizing tool in a first state, where the piston device is moving towards the middle of the piston compartment;
(4) FIG. **3** illustrates a cross sectional side view of the equalizing tool after the first state, where the piston device has been moved to the second end of the piston compartment;
(5) FIG. **4** illustrates a cross sectional side view of the equalizing tool in a second state, where the inner piston section is moving relative to the outer piston section;
(6) FIG. **5** also illustrates the equalizing tool in the second state;
(7) FIG. **6***a* illustrates an enlarged view of the piston in the initial and first states;
(8) FIG. **6***b* illustrates an enlarged view of the piston in the start of the second state;
(9) FIG. **6***c* illustrates an enlarged view of the piston in the end of the second state;
(10) FIG. **6***d* illustrates an enlarged view of the piston in final state;

(11) FIG. **7** illustrates a well tool assembly comprising a plugging and abandonment tool, the downhole equalizing tool and a control and logging system.
(12) Initially, it is referred to FIG. **1**. Here it is shown a well tool assembly **1** comprising three main parts, an upper control and logging system indicated as a dashed rectangle **100**, a lower plugging and abandonment tool indicated as a dashed rectangle **200** and a downhole pressure equalizer **10** connected between the upper control and logging system **100** and the lower plugging and abandonment tool **200**. A central longitudinal axis I-I is indicated in FIG. **1**.
(13) The upper control and logging system **100** typically comprises a wireline interface for connection to a wireline, and a housing in which sensors and control circuitry are provided.
(14) The lower plugging and abandonment tool **200** may be of the type described in WO2013/135583, i.e. comprising a housing **210**, a chamber **220** within the housing **210**, and a heat generating mixture **240** and an igniter **250** located within the chamber **220**.
(15) First, the downhole pressure equalizer **10**, in short referred to as the equalizer **10** will be described in detail. The equalizer **10** comprises an elongated housing **11** in which a piston compartment **12** is provided. The piston compartment **12** has a first or upper end **12***a* provided in fluid communication with an outside OS of the elongated housing **11** and a second end **12***b* provided in fluid communication with the chamber **220** of the well tool **200**.
(16) The housing **11** comprises a longitudinal rod **16** provided centrally through the piston compartment **12** and secured at both ends to the housing **11**. A longitudinal communication bore **14** is provided within the rod **16**. The purpose of the bore **14** is to enable that a sensor located adjacent to the first end **12***a* of the housing **11**, typically a sensor located in the control and logging system **100**, can measure the pressure in the second end **12***b* of the housing **11**, typically the pressure in the chamber **220**.
(17) The relatively long communication bore **14** prevents or at least considerably delays the heat from the heat generation process to damage the pressure sensor.
(18) The equalizer **10** further comprises a piston device **30** slidingly and sealingly engaged within the piston compartment **12**. The piston device **30** is separating the piston compartment **12** into a first sub-compartment **13***a* between the piston device **30** and the first end **12***a* of the housing **11** and a second compartment **13***b* between the piston device **30** and the second end **12***b* of the housing **11**.
(19) It is now referred to FIG. **4**, FIG. **6***a* and FIG. **6***b*. Here it is shown that the piston device **30** comprises an outer piston section **40** and an inner piston section **50**.
(20) The Outer Piston Section **40**
(21) The outer piston section **40** is slidingly and sealingly engaged with the piston compartment **12** and slidingly and sealingly engaged with the outer surface of the rod **16**. The outer piston section **40** comprises a sleeve **41** in which a through bore **42**. Hence, the outer piston section **40** alone does not separate the piston compartment **12** into the two sub-compartments **13***a*, **13***b*. The sleeve **41** further comprises a stop **43***b* in the form of an inwardly protruding restriction of the diameter of the bore **42**. The upper end of the sleeve **41** is also forming a stop indicated as **43***a*.
(22) The inwardly protruding restriction forming the stop **43***b* comprises a bore **46**, allowing fluid from the second compartment **13***b* to enter the lower part of the bore **42**.
(23) Reference number **64***a* is a securing mechanism comprising a transportation element for securing the outer piston section **40** to the housing **11**.
(24) As shown in FIG. **6***b*, there is lower sealing element **64***b* radially outside of the outer piston section **40**, to prevent fluid longitudinal fluid flow radially outside of the outer piston section **40**, i.e. between the outer piston section **40** and the inner surface of the piston compartment **12**.
(25) The Inner Piston Section **50**
(26) The inner piston section **50** is slidingly and sealingly engaged within the through bore **42** of the outer piston section **40** and is also slidingly and sealingly engaged with the outer surface of the rod **16**.
(27) The inner piston section **50** comprises a first or upper end section **53***a*, a second or lower end

section **53***b* and an intermediate section **53***c* longitudinally between the first end section **53***a* and the second end section **53***b*. A first surface **54***a* of the second end section **53***b* is faced towards the first piston compartment **13***a* and a second surface **54***b* of the second end section **53***b* is faced towards the second piston compartment **13***b*.
(28) The second end section **53***b* is sealingly engaged with the bore **42** of the outer piston section **40** by means of a sealing element **65***b*. The second end section **53***b* has a larger outer diameter than the inner diameter of the inwardly protruding restriction of the diameter of the bore **42**. Hence, the second end section **53***b* may be engaged with the stop **43***b* and prevent further downward movement of the inner piston section **50** relative to the outer piston section **40**.
(29) In addition, the first end section **53***a* of the inner piston section **50** is protruding upwardly from the bore **42** and comprises a collar **55** extending radially into contact with the piston compartment **12**. Hence, the first end section **53***a* is slidingly engaged with the piston compartment **12**.
(30) A sliding element **65***a* is provided radially outside of the collar **55**, to reduce friction between the collar **55** and the inner surface of the piston compartment **12**.
(31) The collar **55** comprises bores **56** for allowing fluid to flow from the first compartment **13***a* into the upper part of the bore **42**.
(32) The intermediate section **53***c* comprises a sleeve surrounding the rod **16**.
(33) In FIG. **6***b*, it is further shown that there is a sliding element **66***a* between the first end section **53***a* and the rod **16** and further that there is a sealing element **66***b* between the second end section **53***b* and the rod **16**.
(34) It should be noted that the housing **11**, the outer piston section **40** and the inner piston section **50** are preferably made of aluminium.
(35) Operation of the Equalizer **10**
(36) The operation of the equalizer will now be described in detail.
(37) In FIG. **1**, the equalizer **10** is in an initial state S0, wherein the piston **30** is stationary with respect to piston compartment **12**. The outer piston section **40** and the inner piston section **50** are both located in the first end **12***a* of the piston compartment **12**. The second piston compartment **13***b*, and hence the chamber **220**, is pressurized with a predetermined fluid pressure P0.
(38) The predetermined fluid pressure P0 is typically determined based on the expected well pressure at the well depth at which the downhole pressure equalizer **10** and the well tool **200** is to be used. Hence, the predetermined fluid pressure P0 is larger than the atmospheric pressure at sea level. It should be noted that this will not cause the inner piston section **50** to move relative to the outer piston section **40** as such movement is prevented by the first end **12***a* of the piston compartment **12**.
(39) The equalizer **10** will typically be in this initial state S0 when handled topside and during the initial lowering of the equalizer into the well. The transportation element **64***a* locking the piston **30** to the housing **11** in the position shown in FIG. **1**, thereby preventing relative movement between the outer piston section **40** and the housing **11** and relative movement between the inner piston section **50** and the inner piston section **50**, is removed topside before the operation starts.
(40) In FIG. **2** and FIG. **6***a*, the equalizer **10** is in a first state. The equalizer **10** will typically be in this first state during the lowering of the equalizer **10** into the well.
(41) At some depth, the pressure on the outside OS of the equalizer **10** will increase to a point wherein the pressure difference between the fluid pressure P1 in the first sub-compartment **13***a* and the fluid pressure P2 in the second sub-compartment **13***b* will cause the outer piston section **40** and the inner piston section **50** to move towards the second end **12***b* of the piston compartment **12**, as indicated by arrow A in FIG. **2**. It is not possible for the inner piston section **50** to move downwardly relative to the outer piston section **40**, as the stops **43***a*, **43***b* will prevent relative movement between the inner piston section **30** and the sleeve **41**.
(42) During the movement of the piston **30** in the direction A, the pressure in the second compartment **13***b* will be equalized with the pressure on the outside OS. It should be noted that the

pressure in the second compartment **13***b* will not necessarily be identical to the pressure in the first compartment **13***a*, as it will require a pressure difference above a minimum threshold value before the piston **30** will start moving.

(43) In FIG. **3**, the piston **30** has moved to the second end **12***b* of the piston compartment **12**. It should be noted that this is not a desired situation—as further pressure alignment is not possible. Consequently, a preferred situation is that when the equalizer **10** has reached the desired location in the well, the piston **30** has stopped at a location between the location shown in FIG. **2** and the location shown in FIG. **3**. It should be noted that the expected location the piston after the first state S1 is calculated based on expected well pressure and temperature at the desired location in the well.

(44) The heat generation process may now be started by igniting the heat generation mixture **240** in the chamber **220** by means of the igniter **250**. The heat generation process will increase the pressure inside the chamber **220** and hence also the pressure P2 in the second sub-compartment **13***b*, while the pressure P1 in the first sub-compartment **13***a* will still be equal to the pressure on the outside OS. The equalizer **10** will soon be in its second state S2.

(45) In FIG. **4**, FIG. **5** and FIG. **6***b*, this second state S2 is shown. FIG. **4** shows the second state S2 when the first state S1 ended with the piston **30** having moved to the second end **12***b* of the piston compartment **12** (i.e. the position shown in FIG. **3**), while FIG. **5** shows the second state S2 when the first state S1 ended with the piston **30** having moved to a position somewhere between the positions shown in FIG. **2** and the position shown in FIG. **3**.

(46) The pressure difference between the fluid pressure P1 in the first sub-compartment **13***a* and the fluid pressure P2 in the second sub-compartment **13***b* will now cause the inner piston section **50** to move towards the first end **12***a* of the piston compartment **12** as indicated by arrow B, while the outer piston section **40** is kept stationary relative to the housing **11**.

(47) During the movement of the inner piston section **50** in the direction B, the pressure in the second compartment **13***b* will be equalized with the pressure on the outside OS. It should be noted that the pressure in the second compartment **13***b* will not necessarily be identical to the pressure in the first compartment **13***a*, as it will require a pressure difference above a minimum threshold value before the inner piston section **50** will start moving.

(48) It is now referred to FIG. **6***c*. Here, the inner piston section **50** is almost brought out from the sleeve **41** of the inner piston section **40**. However, the inner piston section **50** is here still considered sealingly engaged with the sleeve **41**.

(49) It is now referred to FIG. **6***d*. Here, the inner piston section **50** has slid out of its sealing engagement within the bore **42** of the outer piston section **40**. The equalizer is now in its final state S3. In the final state S3, the piston device **30** is no longer considered to be a piston, as there is a direct fluid communication between the first sub-compartment **13***a* and the second sub-compartment **13***b.*

(50) It should be noted that additional measures may be taken to ensure that only the inner piston section **50** moves when the equalizer **10** is in the second state S2. This can be done by selecting the number of sealing and/or sliding elements or by selecting properties of the respective sealing and/or sliding elements so that a first friction parameter FP**1**, representing a friction for moving the outer piston section **40** relative to the housing **11**, is larger than a second friction parameter FP**2** representing a friction for moving the inner piston section **50** relative to the outer piston section **40**.

(51) It should further be noted that the pressure development of the heat generation process may develop fast, in some cases the second state S2 will have a duration of a few second, in some cases less than a second.

(52) However, this is considered acceptable, as the heat generation process will also melt the housing **210** and therefore also result in a pressure equalization between the chamber **200** and the outside OS of the housing.

(53) The inner piston section **50** will dampen a substantial part of the first pressure peak resulting from the start of the heat generation process. Hence, the risk of an explosion of the housing **210**

will be considerably reduced.
(54) The above operation may be at least partially be controlled by the control and logging system **100**, which can be configured to verify that the first pressure P1 is equalized with the second pressure P2 when the well tool assembly **1** has been lowered to the desired location in the well and before the heat generation mixture is activated by the igniter **250**. The control and logging system **100** may further be configured to verify that the first pressure P1 is equalized with the second pressure P2 during the heat generation process or after the heat generation process has finished.
(55) The length of the housing **11** of the equalizer **10** may be determined by the expected travel distance for the piston **30** in the first state S1. However, the length of the housing **11** may also be considerably longer than this expected travel length, as the length of the housing **11** is also protecting the sensors etc. in the control and logging system **100** from the heat generation process of plugging and abandonment tool **200**. The length of the outer piston section **40** and the inner piston section **50** may be determined by the expected travel distance for the inner piston section **50** relative to the outer piston section **40**.
(56) However, it is not necessarily cost efficient to tailor-make the equalizer for each operation—a standardized size for many operations may be preferred.
(57) It is now referred to FIG. **1**. Here, a height H12 of the piston compartment **12** is 1581 mm and a height H30 of the piston device **30** is 315 mm. The maximum movement HS1max of the piston device **30** in the first state S1 is therefore equal to the difference between H12 and H30, i.e. 1266 mm.
(58) It is now referred to FIG. **6***c*. Here, the maximum movement HS2max of the inner piston section **50** relative to the outer piston section **40** in the second state S2 is 248 mm. When the inner piston section has moved the maximum movement HS2max, the equalizer **10** will be in its final state S3, in which the inner piston section **50** has slid out of its sealing engagement with the through bore **42** of the outer piston section **40**.
Alternative Embodiments
(59) It should be noted that a wireline may be secured to the upper end of the equalizer **10**, i.e. the well tool assembly **1** only comprises the equalizer **10** and the lower plugging and abandonment tool **200**. It should further be noted that the equalizer **10** may be used together with other well tools.

Claims

1. A downhole pressure equalizer comprising: an elongated housing comprising a piston compartment having a first end provided in fluid communication with an outside of the elongated housing and a second end provided in fluid communication with a process chamber of a well tool; a piston device slidingly and sealingly engaged within the piston compartment, wherein the piston device separates the piston compartment into a first sub-compartment between the piston device and the first end and a second compartment between the piston device and the second end; wherein the piston device comprises: an outer piston section slidingly and sealingly engaged with the piston compartment, wherein the outer piston section comprises a sleeve having a through bore; and an inner piston section slidingly and sealingly engaged within the through bore of the outer piston section, wherein the downhole pressure equalizer is configured to be in one of: a first state, in which a pressure difference between a fluid pressure in the first sub-compartment and a fluid pressure in the second sub-compartment causes the outer piston section and the inner piston section to move towards the second end of the piston compartment; and a second state, in which the pressure difference between the fluid pressure in the first sub-compartment and the fluid pressure in the second sub-compartment causes the inner piston section to move towards the first end of the piston compartment while the outer piston section is kept stationary.
2. The downhole pressure equalizer according to claim 1, wherein the downhole pressure equalizer is configured to be in an initial state before being in the one of the first state and the second state,

wherein in the initial state, the outer piston section and the inner piston section are located in the first end of the piston compartment, and the second piston compartment is pressurized with a predetermined fluid pressure.

3. The downhole pressure equalizer according to claim 1, wherein the downhole pressure equalizer is configured to be in a final state after being in the one of the first state and the second state, wherein in the final state, the inner piston section has slid out of its sealing engagement with the through bore of the outer piston section.

4. The downhole pressure equalizer according to claim 1, further comprising a longitudinal communication bore for sensing the fluid pressure in the second sub-compartment at a location adjacent to the first end of the piston compartment.

5. The downhole pressure equalizer according to claim 1, further comprising a longitudinal rod provided through the piston compartment and secured at both ends to the elongated housing, wherein the outer piston section is slidingly and sealingly engaged with an outer surface of the longitudinal rod, and wherein the inner piston section is slidingly and sealingly engaged with the outer surface of the longitudinal rod.

6. The downhole pressure equalizer according to claim 5, wherein the longitudinal communication bore is provided within the longitudinal rod.

7. The downhole pressure equalizer according to claim 1, wherein a first friction parameter representing a friction for moving the outer piston section relative to the elongated housing is larger than a second friction parameter representing a friction for moving the inner piston section relative to the outer piston section.

8. The downhole pressure equalizer according to claim 7, wherein the first friction parameter and the second friction parameter are defined by the number of, or properties of, sealing elements sealingly engaged between the piston device and the piston compartment and between the inner piston section and the outer piston section.

9. The downhole pressure equalizer according to claim 1, wherein the outer piston section comprises a sleeve in which the through bore is provided, wherein the sleeve comprises a stop preventing relative movement between the inner piston section and the sleeve when the downhole pressure equalizer is in the first state.

10. The downhole pressure equalizer according to claim 1, wherein the inner piston section comprises a first end section, a second end section and an intermediate section, wherein the second end section is sealingly engaged within the through bore of the outer piston section, wherein a first surface of the second end section is faced towards the first piston compartment, and wherein a second surface of the second end section is faced towards the second piston compartment.

11. The downhole pressure equalizer according to claim 10, wherein the first end section of the inner piston section comprises a collar protruding from the through bore, wherein the collar is slidingly engaged with the piston compartment.

12. A well tool assembly for forming a permanent barrier in a well, wherein the well tool assembly comprises: a plugging and abandonment tool comprising a housing, a chamber within the housing, and a heat generating mixture and an igniter located within the chamber; and the downhole pressure equalizer according to claim 1 connected above the plugging and abandonment tool, wherein the second sub-compartment is provided in fluid communication with the chamber of the plugging and abandonment tool.

13. Well The well tool assembly according to claim 12, further comprising: a control and logging system connected above the downhole pressure equalizer, wherein the control and logging system comprises: a first pressure sensor for logging a parameter representative of a first pressure in the first sub-compartment; and a second pressure sensor for logging a parameter representative of a second pressure in the second sub-compartment, wherein the control and logging system is configured to verify that the first pressure is equalized with the second pressure when the well tool assembly has been lowered to a desired location in the well and before the heat generation mixture

is ignited by the igniter.
14. The well tool assembly according to claim 13, wherein the control and logging system is configured to verify that the first pressure is equalized with the second pressure during a heat generation process or after the heat generation process has finished.